Chapter 1 - Introduction To Waves - Notes

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1 Introduction to Waves

A **wave** is any recognizable signal that is transferred from one part of a medium to another with a recognizable velocity of propagation. A medium is a thing a wave passes through, such as water or air.

A wave is caused by a disturbance that disrupts the medium from its normal resting state. Think of a rock hitting water or someone speaking. That is what causes the wave to begin propagation.

When observing a wave, one should always be able to track a signal. The **signal** is the specific, measurable feature of that disturbance that you can track as it travels through the medium.

1.1 Examples of Waves

One good example of a wave is a pebble hitting a pond. When a pebble hits a pond, it causes a disturbance. This disturbance creates a signal, in this case, crests (waves) in the pond. This waves propagate outwards from the center of where the pebble hit with a recognizable velocity.

Another good example is car traffic. When a car stops at a light, it creates a "disturbance" in traffic. All the cars behind that car will stop. We can track the end of the line as our signal. The more cars that get added to the line, the further back our signal moves.

Wave Phenomena	**Medium**	**Signal**
Ripples on a pond	Surface layer of water	Extreme water surface height
Compression waves	Elastic bar	Maximum longitudinal displacement
Sound waves	Gas or liquid	Pressure extrema
Shock waves	Gas	Abrupt change in pressure
Traffic waves	Car traffic on a road	Abrupt change in traffic density (hitting the end of the line)
Epizootic waves	A population susceptible to contracting a disease	Extreme number of infectives